

CMAN :=

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Task 1 :=

$$N = 512$$

Each multiply then add pair = 2 Mac operations

a) $C_{\text{dense}} = 2 \times N^2 = 2 \times 512^2 = 524,288 \text{ FLOPs}$

b) Dense memory bytes :=

Here $\text{FP32} = 4 \text{ bytes}$

$$M_{\text{dense}} = 4 \times N^2 = 4 \times 512^2 = 1,048,576 \text{ bytes}$$

c) Sparse compute :=

$C_{\text{dense}} \times (1-s)$ $\rightarrow$ only remaining fraction $(1-s)$ weights

$$C_{\text{sparse}} = 2N^2(1-s)$$
$$= 2 \times (512)^2 (1-s)$$

$$C_{\text{sparse}} = 524,288 (1-s) \text{ FLOPs}$$

d) Sparse memory :=

CSR format stores 3 things.

i) The actual non-zero values :=

Each one is $\text{FP32} = 4 \text{ bytes}$

$$4 \times N^2 (1-s) \text{ bytes}$$

ii) The column index for each non-zero

$$\text{column index} = 4 \times N^2 (1-s) \text{ bytes}$$

iii) The row pointer array:

(2)

$N+1$ entries, each $\text{INT32} = 4$ bytes.

$4 \times (N+1)$ bytes

$$M_{\text{sparse}} = 4 \times N^2(1-S) + 4 \times N^2(1-S) + 4 \times (N+1)$$

$$M_{\text{sparse}} = 8N^2(1-S) + 4(N+1) \text{ bytes}$$

Task 2: =

$$S_{\text{FLOPs}} = \frac{C_{\text{dense}}}{C_{\text{sparse}}} = \frac{2N^2}{2N^2(1-S)} = \frac{1}{1-S}$$

$$2x = ?$$

$$\frac{1}{1-S} = 2 \Rightarrow 1-S = \frac{1}{2}$$

$$S = 0.5 = 50$$

$$S = 50\%$$

Task 3: =

$$M_{\text{sparse}} = M_{\text{dense}}$$

$$8N^2(1-S) + 4(N+1) = 4N^2$$

$$8N^2 - 8N^2S + 4N + 4 = 4N^2$$

$$-8N^2S = 4N^2 - 8N^2 - 4N - 4 \quad \text{Divide both sides by } -8N^2:$$

$$S = \frac{4N^2 + 4N + 4}{8N^2}$$

$$S = \frac{1}{2} + \frac{1}{2N} + \frac{1}{2N^2}$$

(3)

$$N = 512$$

$$S = \frac{1}{2} + \frac{1}{1024} + \frac{1}{2 \times 512^2} = \frac{1}{2} + \frac{1}{1024} + \frac{1}{524288}$$

$$S = 0.5 + 0.000976 + 0.0000019$$

$$S \approx 0.500978$$

$$S_{\text{break even}} = 50.1$$

That tiny extra 0.1% come from row pointer array those 513 extra integers that CSR always has to store no matter what even if the matrix is almost entirely zero. If the row pointer array didn't exist the break even would be exactly 50%.

Task 4 :=

$$S = 0.9, N = 512, \text{ Bandwidth} = 320 \text{ GB/s}$$

Dense time :=

$$M_{\text{dense}} = 4 \times 512^2 = 1,048,576 \text{ bytes}$$

$$T_{\text{dense}} = \frac{1,048,576}{320 \times 10^9} \approx 3.28 \text{ MS}$$

Sparse time :=

first count the non-zeros:

$$N^2 \times (1 - S) = 512^2 \times (1 - 0.9) = 262,144 \times 0.1 \\ = 26214.4 \text{ Non zeros}$$

Now compute each part of sparse memory
Values = $26,214 \times 4 = 104,858 \text{ bytes}$

$$\text{Column index} = 26,214 \times 4 = 104,858 \text{ bytes}$$

$$\text{Row pointers} = 513 \times 4 = 2052 \text{ bytes}$$

$$M_{\text{sparse}} = 104,858 + 104,858 + 2052 = 211,768$$

$$T_{\text{sparse}} = \frac{211,768}{320 \times 10^9} \approx 0.662 \text{ ns}$$

Final Speedup: =

$$S = \frac{T_{\text{dense}}}{T_{\text{sparse}}} = \frac{3.28}{0.662} \approx 4.95 \times$$

$$\boxed{S \approx 4.95 \times}$$